Detailed Mathematical and Logical Review

of "Informativity and Identifiability for Identification of Networks of Dynamical Systems"

Technical review for internal research discussion

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Manuscript reviewed. A. Hansson, J. V. Galvão da Mata, and M. S. Andersen, "Informativity and Identifiability for Identification of Networks of Dynamical Systems," arXiv:2602.20107v2, 26 February 2026. The theorem, equation, example, and page references below refer to that version.

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1 Overall assessment

The paper develops an appealing program: use structural rank and Gröbner-basis calculations to study two distinct questions in dynamic-network identification, namely informativity of predictor inputs and generic local identifiability of open-loop modules. The block formulation is flexible, the use of algebraic geometry is potentially valuable, and several of the examples contain useful ideas.

The present version is nevertheless not mathematically correct as stated. The principal difficulties are theorem-level rather than merely expository:

1. Theorem 1 replaces a condition on the *true* transfer matrix by a generic-rank condition, but then concludes informativity for the actual experiment. Generic full rank only excludes a proper algebraic exceptional set; it does not exclude the true system from that set.
2. Theorem 2's vertex-disjoint-path characterization is false for the paper's advertised class of arbitrary generically invertible structured matrices P. The theorem implicitly identifies the ith row and ith column of P as one state vertex, although an arbitrary P need not possess the corresponding distinguished matching.
3. Theorem 3 does not consistently distinguish an admissible rank-drop set from its Zariski closure, omits saturation by all well-posedness denominators, and in part 3 confuses the rank-$< k$ locus with the locus below generic rank.
4. Theorem 4 contains the right generic fiber-dimension idea, but its literal "any admissible G_c" statement is false. The image of a rational map need not be an affine variety, and equal dimensions yield zero-dimensional fibers only over a generic subset of the image closure.

There are additional issues concerning the passage between rational transfer functions and evaluation at one frequency, selection of only some predictor columns, and several claims in the examples. Most of the algebraic-geometry program can be repaired, but the graph theorem must either be restricted substantially or replaced by a bipartite matching statement.

1.1 Status summary

Claim or component	Assessment	Main point
Predictor formula and recovery of G_c	Correct with assumptions	The displayed algebra is correct when the required rational inverses exist.
Parseval argument	Repairable	The frequency integral and conjugate transpose are missing; positivity must be used under the integral.
Theorem 1	False as stated	Generic full rank does not imply full rank at the true parameter.
Block-rank identity	Correct	On the well-posed domain, $\operatorname{rank} M = n + \operatorname{rank} \Pi$.
Theorem 2	False in the stated generality	An arbitrary invertible P need not support the same-index row–column contraction used by the directed graph.
Theorem 3	Substantially incomplete	Saturation and closure must be stated correctly, and k must equal the generic rank before the exceptional locus is identified.
Theorem 4	Correct generic idea, false literal quantifier	Equal dimensions characterize generic zero-dimensional fibers, not every fiber.
Subnetwork algebra	Largely correct	It inherits the qualifications required for Theorems 1–4.
Examples 1–5	Mixed	Several conclusions are correct, but Example 1's transfer matrix is misprinted, Example 2 relies on an invalid column-selection argument, and Example 5 is stronger than reported.

The appropriate recommendation is **major revision**.

2 Predictor algebra and spectral preliminaries

2.1 Predictor formulas

The paper starts from

$$P(q)w_k = Q(q)u_k, \qquad z_k = R(q)w_k + S(q)u_k, \tag{1}$$

partitions $u_k = (r_k, e_k)$ and $z_k = (y_k, r_k)$, and writes the measured-output dynamics as

$$y_k = G_c(q)r_k + H_c(q)e_k. \tag{2}$$

The innovation representation is

$$\widehat{y}_k = G_c(q)r_k + G_o(q)\varepsilon_k, \qquad y_k = \widehat{y}_k + \varepsilon_k. \tag{3}$$

Eliminating ε_k gives

$$\widehat{y}_k = W(q)z_k, \qquad W = (I + G_o)^{-1}\begin{bmatrix} G_o & G_c \end{bmatrix}. \tag{4}$$

Consequently,

$$W_y = (I + G_o)^{-1} G_o, \qquad W_r = (I + G_o)^{-1} G_c, \tag{5}$$

and hence

$$G_c = (I - W_y)^{-1} W_r. \tag{6}$$

This algebra is correct. The manuscript should state explicitly the rational invertibility assumptions used for $I + G_o$ and $I - W_y$, in addition to stability of the resulting predictor.

The innovation assumptions also need sharpening. Independence, identical distribution, and zero mean do not by themselves express the defining one-step prediction property. For a prediction-error interpretation, the paper should specify the relevant information set and state either the innovation property or the corresponding orthogonality conditions.

2.2 Correct Parseval argument

Let two stable candidate predictors be denoted by W_1 and W_2, and set

$$\Delta W = W_1 - W_2, \qquad d_k = \Delta W(q) z_k. \tag{7}$$

Under the standard second-order stationarity assumptions,

$$\mathbb{E}[d_k d_k^*] = \frac{1}{2\pi} \int_{-\pi}^{\pi} \Delta W(e^{i\omega}) \Phi_z(\omega) \Delta W(e^{i\omega})^* \, \mathrm{d}\omega. \tag{8}$$

The integrand is positive semidefinite. Therefore, if the left-hand side is zero, then the integrand is zero for almost every ω. If $\Phi_z(\omega) \succ 0$ almost everywhere, it follows that

$$\Delta W(e^{i\omega}) = 0 \quad \text{for almost every } \omega. \tag{9}$$

The manuscript instead displays the integrand itself as equal to zero without writing the integral. The intended reasoning is valid after restoring (8) and using a conjugate transpose rather than an ordinary transpose, unless a special real-frequency convention is introduced.

3 The informativity theorem

3.1 The fixed-system implication

Let θ_0 denote the true system and

$$\Pi(q; \theta_0) = R(q; \theta_0) P(q; \theta_0)^{-1} Q(q; \theta_0) + S(q; \theta_0) \tag{10}$$

be the transfer matrix from $u = (r, e)$ to z. Then

$$\Phi_z(\omega) = \Pi(e^{i\omega}; \theta_0) \Phi_u(\omega) \Pi(e^{i\omega}; \theta_0)^*. \tag{11}$$

Thus, if $\Phi_u(\omega) \succ 0$ and $\Pi(e^{i\omega}; \theta_0)$ has full row rank almost everywhere, then $\Phi_z(\omega) \succ 0$ almost everywhere. Together with (8), this proves informativity in the paper's Definition 1.

This is a statement about the *actual* matrix $\Pi(\cdot; \theta_0)$. The manuscript itself observes this immediately before Theorem 1. The subsequent replacement of actual rank by generic rank is where the logical gap occurs.

3.2 Counterexample to Theorem 1

Proposition 3.1 (Generic full rank does not imply actual informativity)**.** *There are stable systems in the paper's model class for which* Π *has full generic row rank and* $\Phi_u \succ 0$*, but the resulting signal* z *is not informative in the sense of Definition 1.*

Proof. Let $L = q^{-1}$ and consider one measured dynamic node together with one measured external signal r. Choose

$$P = 1 - aL, \qquad Q_r = 0, \qquad Q_e = 1, \qquad R_y = 1, \qquad S_{yr} = 0, \qquad S_{ye} = s. \tag{12}$$

The transfer from $u = (r, e)$ to $z = (y, r)$ is

$$\Pi(a, s) = \begin{bmatrix} 0 & \dfrac{1}{1 - aL} + s \\ 1 & 0 \end{bmatrix}. \tag{13}$$

As a rational matrix in the free entries a and s, this has generic rank two.

Fix any a_0 with $|a_0| < 1$ and choose the true stable feedthrough filter

$$s_0 = -\frac{1}{1 - a_0 L}. \tag{14}$$

The two paths from e to y then cancel exactly, so the true experiment satisfies

$$y \equiv 0, \qquad z = (0, r). \tag{15}$$

This remains true even when the joint spectrum of (r, e) is positive definite.

Now consider two stable candidate models with $s = 0$ and distinct parameters $a_1 \neq a_2$. For these candidates, the ordinary one-step predictor is

$$\widehat{y}_k = a_i L y_k, \qquad i = 1, 2, \tag{16}$$

so their predictor matrices have different y-columns. On the true data, however, $y \equiv 0$, and both predictors produce the same output. Hence equality of predicted outputs does not imply equality of predictor transfer matrices. The signal z is therefore not informative. □

The counterexample is nongeneric by construction. That is precisely why it refutes the theorem: a generic-rank hypothesis cannot rule out the true parameter belonging to the rank-drop variety.

3.3 Corrected form of Theorem 1

The result should be separated into an actual-system theorem and a generic corollary.

Theorem 3.1 (Fixed-system informativity)**.** *Assume that the candidate predictors are stable and that all signals admit the spectra used in* (8)*. If*

$$\Phi_u(\omega) \succ 0 \quad \textit{and} \quad \operatorname{rank} \Pi(e^{i\omega}; \theta_0) = \dim z \tag{17}$$

for almost every ω*, then* z *is informative in the sense of Definition 1.*

Theorem 3.2 (Generic informativity)**.** *Let the admissible parameter set be an irreducible algebraic or quasi-affine variety, and let* $\Pi(z;\theta)$ *be rational and regular on a nonempty well-posed open subset. If* Π *has full generic row rank, then there is a proper algebraic exceptional subset of parameters such that, for every true parameter outside that subset,* $\Pi(e^{i\omega};\theta_0)$ *has full row rank for almost every* ω*. If in addition* $\Phi_u \succ 0$ *almost everywhere, informativity holds for those generic true systems.*

The conclusion is therefore "informativity holds generically," not "the observed z is informative" for every admissible true system. All later applications of Theorem 1 should carry this qualifier.

3.4 Identifying only selected predictor columns

Remark 1 proposes that, when only selected columns of W are of interest, the remaining rows of the matrix M can simply be removed. This is not valid unless the unselected predictor columns are fixed across candidate models or an additional conditional excitation condition is imposed.

Write

$$W = \begin{bmatrix} W_s & W_n \end{bmatrix}, \qquad z = \begin{bmatrix} z_s \\ z_n \end{bmatrix}. \tag{18}$$

Equality of two predicted outputs gives

$$\Delta W_s z_s + \Delta W_n z_n = 0. \tag{19}$$

A change in the selected columns can be cancelled by a change in the nuisance columns. For example, if $z_2 = z_1$, then

$$W = \begin{bmatrix} 0 & 0 \end{bmatrix}, \qquad \widetilde{W} = \begin{bmatrix} D & -D \end{bmatrix}, \qquad D \neq 0, \tag{20}$$

produce identical predictions although their first columns differ.

A valid reduced-column result can be obtained under any of the following kinds of assumptions:

- the nuisance columns are known and therefore satisfy $\Delta W_n = 0$;
- the model class is redefined so that the omitted columns cannot occur;
- a conditional spectral condition is imposed on the innovation in z_s after linear prediction from z_n.

For instance, when Φ_{nn} is nonsingular, the relevant residual spectrum is the Schur complement

$$\Phi_{s|n} = \Phi_{ss} - \Phi_{sn}\Phi_{nn}^{-1}\Phi_{ns}. \tag{21}$$

Positive definiteness of $\Phi_{s|n}$ can rule out cancellation in (19); positive definiteness of Φ_{ss} alone does not.

4 Generic rank and the graph-theoretic theorem

4.1 The block-rank identity is correct

For

$$M = \begin{bmatrix} P & Q \\ -R & S \end{bmatrix}, \qquad \Pi = RP^{-1}Q + S, \tag{22}$$

the manuscript uses the factorization

$$\begin{bmatrix} I & 0 \\ RP^{-1} & I \end{bmatrix} M \begin{bmatrix} I & -P^{-1}Q \\ 0 & I \end{bmatrix} = \begin{bmatrix} P & 0 \\ 0 & \Pi \end{bmatrix}. \tag{23}$$

Both multiplying matrices are invertible whenever P is invertible. Consequently,

$$\operatorname{rank} M = n + \operatorname{rank} \Pi. \tag{24}$$

This identity is sound and is the natural bridge to structural matching methods.

4.2 Counterexample to Theorem 2

The paper constructs a directed graph by identifying the ith row and the ith column of the P block with one vertex W_i. It then states that the maximum number of vertex-disjoint paths from U to Z equals $\operatorname{rank}_{\mathrm{g}} \Pi$. For an arbitrary invertible P, that row–column identification is unjustified.

Proposition 4.1 (Theorem 2 is false for arbitrary invertible P)**.** *Let*

$$P = \begin{bmatrix} 0 & a \\ b & 0 \end{bmatrix}, \qquad Q = \begin{bmatrix} q \\ 0 \end{bmatrix}, \qquad R = \begin{bmatrix} r & 0 \end{bmatrix}, \qquad S = 0, \tag{25}$$

where a, b, q, r are algebraically independent. Then the paper's directed graph contains a path from U to Z, but Π is identically zero.

Proof. The matrix P is generically invertible and

$$P^{-1} = \begin{bmatrix} 0 & b^{-1} \\ a^{-1} & 0 \end{bmatrix}. \tag{26}$$

Therefore

$$\Pi = RP^{-1}Q = 0, \qquad \operatorname{rank}_{\mathrm{g}} \Pi = 0. \tag{27}$$

On the other hand, $Q_{1,1} = q$ creates the edge $U \to W_1$, and $R_{1,1} = r$ creates the edge $W_1 \to Z$ under the paper's definition. Thus its graph contains the path

$$U \longrightarrow W_1 \longrightarrow Z, \tag{28}$$

which would predict generic rank one. □

The failure is structural. In (25), nonsingularity of P is supplied by the off-diagonal perfect matching

$$W_1^{\mathrm{col}} \leftrightarrow W_2^{\mathrm{row}}, \qquad W_2^{\mathrm{col}} \leftrightarrow W_1^{\mathrm{row}}, \tag{29}$$

not by the same-index row–column pairs contracted in the manuscript's graph. The classical state-space result for $sI - A$ has a distinguished diagonal matching supplied by the sI term; an arbitrary structured P does not.

4.3 Correct general matching statement

Assume for the moment that all structural nonzeros are algebraically independent. Let $\mathcal{B}(M)$ be the bipartite graph having one vertex for each row of M, one vertex for each column of M, and an edge for every structural nonzero. The standard structural-rank theorem gives

$$\operatorname{rank}_{\mathrm{g}} M = \nu(\mathcal{B}(M)), \tag{30}$$

where ν is the maximum matching cardinality. Combining this with (24) yields the exact statement

$$\boxed{\operatorname{rank}_{\mathrm{g}} \Pi = \nu(\mathcal{B}(M)) - n.} \tag{31}$$

A directed path formulation can be recovered only after a perfect matching of the P block has been selected and the corresponding row–column pairs are contracted. The resulting auxiliary digraph is matching-dependent. For matrices of the special form $P = I - G$, the diagonal identity entries provide the required canonical matching, which explains why a same-index path theorem is natural in much of the dynamic-network literature. The present manuscript explicitly allows more general P, so it cannot use that contraction without an additional hypothesis.

4.4 Fixed entries and algebraic dependencies

Remark 3 correctly notes that the path result assumes all nonzero entries to be free. This qualification has substantial consequences:

- Fixed nonzero entries are not algebraically independent parameters.
- Repeated transfer functions and rational constraints can create cancellations that ordinary structural matching does not detect.
- Once dependencies are admitted, even (31) need not equal the actual generic rank in the constrained parameterization; direct polynomial or rational rank analysis is then required.

Several examples contain fixed entries equal to one. Their ranks may of course still be computed directly, but they should not be presented as automatic consequences of the fully free structural theorem.

5 The Gröbner-basis rank calculation

5.1 The admissible parameter domain is quasi-affine

The transfer matrix Π is rational and is defined only where the relevant denominators are nonzero. At a minimum, the well-posed domain excludes

$$\det P = 0. \tag{32}$$

Additional denominators arise from transfer-function parameterizations, rational constraints, and row-wise clearing of denominators. Consequently, the natural parameter domain is not generally an affine variety but a quasi-affine set

$$X^{\circ} = X \setminus \mathcal{V}(\Delta), \tag{33}$$

where X is the algebraic constraint variety and Δ is a product containing every denominator whose nonvanishing is required.

The admissible rank-drop set is relatively closed in X°, but need not be closed in the ambient affine space. For example,

$$\Pi(x, y) = \frac{y}{x}, \qquad x \neq 0, \tag{34}$$

has admissible rank-zero set

$$\{(x, y) : y = 0,\ x \neq 0\}. \tag{35}$$

Its Zariski closure is the full line $y = 0$, including the inadmissible point $x = 0$. Thus the manuscript's assertion that the actual rank-drop set is a proper affine variety is not literally correct.

5.2 Saturation is the correct algebraic operation

Let $I_X \subset K[x_1, \ldots, x_N]$ describe all polynomial parameter constraints, where K is the chosen coefficient field. Let $\widetilde{\Pi}$ be a polynomial matrix obtained by clearing denominators, and let $I_k(\widetilde{\Pi})$ denote the ideal generated by its $k \times k$ minors. Let

$$\Delta = (\det P) \prod_j d_j \tag{36}$$

contain all well-posedness and parameterization denominators. The appropriate saturated ideal is

$$J_k = (I_X + I_k(\widetilde{\Pi})) : \Delta^\infty. \tag{37}$$

Then

$$\{x \in X^\circ : \operatorname{rank} \Pi(x) < k\} = \mathcal{V}(J_k) \cap X^\circ, \tag{38}$$

and $\mathcal{V}(J_k)$ is its Zariski closure in X.

Introducing a new variable t and the equation $1 - t\Delta = 0$ is a standard way to compute the same saturation:

$$J_k = \langle I_X, I_k(\widetilde{\Pi}), 1 - t\Delta \rangle \cap K[x_1, \ldots, x_N]. \tag{39}$$

A lexicographic elimination basis therefore computes the closure $\mathcal{V}(J_k)$. It does not remove from that closure all limit points satisfying $\Delta = 0$; exact admissibility is recovered by intersecting with X° as in (38).

5.3 Problems in Theorem 3

The three clauses of Theorem 3 need to be reformulated.

Clause 1. If the saturated ideal is the unit ideal, then the admissible rank-$< k$ set is empty, so every admissible parameter has rank at least k. This part is essentially correct, provided Δ includes all denominators and all constraints are included.

Clause 2. On an irreducible reduced parameter variety, the elimination ideal being zero in the coordinate ring means that all $k \times k$ minors vanish identically in the function field, so the generic rank is below k. With constraints, the correct comparison is $J_k = I_X$ in the ambient polynomial ring, not necessarily a literal zero basis. For reducible X, the generic rank can differ between components and should be analyzed componentwise.

Clause 3. For a specified k, the minors define the locus

$$\operatorname{rank} \Pi < k. \tag{40}$$

They define the locus below generic rank only after one has established

$$k = \operatorname{rank}_{\mathrm{g}} \Pi. \tag{41}$$

If the generic rank is three and $k = 2$, the 2×2 minors detect rank at most one and miss all points of rank two. Thus the last sentence of clause 3 is false for arbitrary k. Moreover, the elimination ideal defines the closure of the admissible locus, not the exact quasi-affine set.

5.4 Corrected rank algorithm

A rigorous replacement is as follows.

1. Specify the algebraic constraint variety $X = \mathcal{V}(I_X)$ and the denominator product Δ.
2. Form J_k as in (37) for successive k.
3. On each irreducible component, determine the largest k for which at least one $k \times k$ minor is nonzero in the component's function field. This is the componentwise generic rank r.
4. The admissible exceptional set on that component is

$$\mathcal{V}(J_r) \cap X^{\circ}, \tag{42}$$

and its affine closure is $\mathcal{V}(J_r)$.

Rational equations cannot simply be inserted without qualification. Their denominators must first be cleared and included in Δ.

6 Generic local network identifiability

6.1 The image need not be an affine variety

The identifiability map is

$$F(X) = R_y P^{-1} Q_r + S_{yr}. \tag{43}$$

The manuscript denotes the well-posed domain by V_o and the image by $V_c = F(V_o)$. Even when F is regular on V_o, the image $F(V_o)$ is generally only constructible; it need not be Zariski closed. The geometric object whose dimension is computed by elimination is

$$Y = \overline{F(V_o)}. \tag{44}$$

A basic example already occurs in the manuscript:

$$F(g, q) = (g, gq). \tag{45}$$

Its image is

$$\{(a, b) : a \neq 0\} \cup \{(0, 0)\}, \tag{46}$$

which is not closed, whereas its closure is all of $\mathbb{C}^2$. Remark 8 effectively computes the smallest variety containing the image, but the theorem and surrounding terminology should consistently use the closure.

6.2 The literal quantifier in Theorem 4 is false

Theorem 4 states generic local network identifiability "for any admissible G_c" when $\dim V_o = \dim V_c$. The proof only invokes a generic fiber-dimension statement, and the stronger quantifier is false.

For (45), the domain and image closure both have dimension two. If $a \neq 0$, the inverse is unique:

$$g = a, \qquad q = \frac{b}{a}. \tag{47}$$

At the admissible image point $(a, b) = (0, 0)$, however,

$$F^{-1}(0, 0) = \{(0, q) : q \in \mathbb{C}\}, \tag{48}$$

which is one-dimensional. Thus equal dimensions do not imply local identifiability at every admissible image point. This exceptional fiber is also present in the paper's Example 1.

6.3 Correct generic fiber theorem

A suitable replacement is the following standard statement.

Theorem 6.1 (Generic local identifiability)**.** *Let X be an irreducible quasi-affine variety over $\mathbb{C}$, let $F : X \to \mathbb{C}^N$ be regular, and define*

$$Y = \overline{F(X)}. \tag{49}$$

Then there exists a nonempty Zariski-open subset $Y^\circ \subseteq Y$ such that

$$\dim F^{-1}(y) = \dim X - \dim Y \qquad \text{for all } y \in Y^\circ. \tag{50}$$

Consequently, the generic fibers are zero-dimensional if and only if

$$\dim X = \dim Y. \tag{51}$$

In that case every point of a generic set-theoretic fiber is isolated, so the parameter is locally identifiable up to a finite discrete ambiguity for generic y.

For a reducible parameter set, this must be applied to each irreducible component that is intended to represent admissible systems. A comparison of only the maximum dimensions of the union can hide a component with systematically larger fibers.

6.4 A cheaper Jacobian certificate

In characteristic zero, the image dimension equals the generic rank of the differential at a generic smooth point. If X is an open subset of $\mathbb{C}^k$, then

$$\dim Y = \operatorname{rank}_{\mathrm{g}} J_F(X), \tag{52}$$

where J_F is the Jacobian of the scalar entries of F with respect to the k parameters. Thus generic local identifiability is equivalent to

$$\operatorname{rank}_{\mathrm{g}} J_F = k. \tag{53}$$

For constrained parameter varieties, the Jacobian should be restricted to the tangent space of X.

This provides a direct exact certificate: it is enough to exhibit one admissible point at which a $k \times k$ Jacobian minor is nonzero. For several examples in the paper, this is much simpler than computing an elimination Gröbner basis.

6.5 Transfer functions versus one-frequency values

The manuscript alternates between two different mathematical interpretations:

1. the entries of X are rational transfer functions and are manipulated symbolically;
2. one fixes a frequency $z = z_0$, regards $X(z_0) \in \mathbb{C}^k$, and uses an ordinary Euclidean norm in the definition of local identifiability.

Pointwise equality at one frequency does not identify an arbitrary rational transfer function. Infinitely many distinct rational functions take the same value at z_0. A rigorous paper should choose one of the following frameworks.

- **Symbolic field framework:** work algebraically over $K = \mathbb{C}(z)$, treating the unknown module values as coordinates over that field. Then the geometric statement is algebraic, but the Euclidean neighborhood language at one frequency should be removed or translated carefully.
- **Finite-dimensional coefficient framework:** parameterize every transfer function by finitely many coefficients of prescribed order and use sufficiently many frequency values or time-domain coefficients.
- **Pointwise framework:** explicitly state that only the complex module values at a fixed generic frequency are being identified; no immediate transfer-function conclusion should then be claimed.

Saying that consideration of one frequency entails no limitation requires a theorem connecting the chosen pointwise statement to equality of rational functions under the paper's parameterization. That theorem is not supplied in the present version.

6.6 Identifiability of selected modules in the presence of nuisance parameters

Suppose the parameter vector is partitioned as

$$X = (\tau, \eta), \tag{54}$$

where τ contains the modules of interest and η contains nuisance modules. Local identifiability of τ does not require the full fiber of F to be zero-dimensional; it requires the projection of the fiber onto the τ coordinates to be locally a singleton. At a regular generic point, the differential condition is

$$v \in \ker J_F(X) \quad \Longrightarrow \quad v_\tau = 0. \tag{55}$$

Equivalently, all infinitesimal nonidentifiable directions must lie entirely in nuisance coordinates. This is the appropriate local criterion behind any claim that fewer predictor equations can still identify selected open-loop modules. Simple deletion of equations or predictor columns does not by itself establish (55).

7 Subnetwork constructions

The algebraic partition of the network into A and B parts is useful. From

$$P_A w_k^A = \begin{bmatrix} -P_{AB} & Q_{Ar} \end{bmatrix} \begin{bmatrix} w_k^B \\ r_k \end{bmatrix} + Q_{Ae} e_k, \tag{56}$$

$$y_k = R_{yA} w_k^A + \begin{bmatrix} R_{yB} & S_{yr} \end{bmatrix} \begin{bmatrix} w_k^B \\ r_k \end{bmatrix} + S_{ye} e_k, \tag{57}$$

the corresponding boundary-to-output map is

$$R_{yA} P_A^{-1} \begin{bmatrix} -P_{AB} & Q_{Ar} \end{bmatrix} + \begin{bmatrix} R_{yB} & S_{yr} \end{bmatrix}. \tag{58}$$

This substitution is correct when w^B is actually measured and treated as an external predictor input.

The alternative construction based on measuring

$$\begin{bmatrix} \widetilde{w}^B \\ \widehat{w}^B \end{bmatrix} = \begin{bmatrix} -P_{AB} \\ R_{yB} \end{bmatrix} w^B \tag{59}$$

is also algebraically correct if those filtered interconnection signals are physically available. They cannot generally be constructed from a measurement of w^B alone without already knowing the unknown filters P_{AB} and R_{yB}.

The broad conclusion that arbitrary subnetworks can then be analyzed by the preceding theorems must inherit all corrections above:

- full generic rank gives only generic informativity;
- the directed path theorem requires a distinguished matching or must be replaced by bipartite matching;
- rational rank loci require saturation;
- targeted module identifiability must account for nuisance directions as in (55).

8 Audit of the examples

8.1 Example 1: transfer matrix, determinant, and exceptional fiber

Let

$$d = 1 - G_1 G_2. \tag{60}$$

Solving the four node equations in Example 1 gives the transfer from (r, e_1, e_2) to (w_1, w_3, r) as

$$\Pi = \begin{bmatrix} \dfrac{G_1(Q_1 + G_2 Q_2)}{d} & \dfrac{H_1}{d} & \dfrac{G_1 H_2}{d} \\ \dfrac{G_2(G_1 Q_1 + Q_2)}{d} & \dfrac{G_2 H_1}{d} & \dfrac{H_2}{d} \\ 1 & 0 & 0 \end{bmatrix}. \tag{61}$$

Therefore

$$\det \Pi = \frac{H_1 H_2}{1 - G_1 G_2}. \tag{62}$$

The displayed matrix in the manuscript omits the factors H_1, H_2, and its stated determinant has the wrong sign. If H_1 and H_2 are fixed nonzero noise filters, the intended full-rank conclusion remains correct under well-posedness. If they are allowed to vanish, well-posedness alone is insufficient.

The selected closed-loop map is

$$F(G_1, Q_1) = (G_1, G_1 Q_1). \tag{63}$$

For $G_c^1 \neq 0$, it has the unique inverse

$$G_1 = G_c^1, \qquad Q_1 = \frac{G_c^2}{G_c^1}. \tag{64}$$

Thus the example establishes generic global identifiability, not merely generic local identifiability. At $G_c = (0, 0)$ the fiber is $\{(0, Q_1)\}$ and is positive-dimensional. This is exactly the exceptional fiber showing why Theorem 4 cannot contain an "any admissible G_c" quantifier.

The sentence saying that the reason the example does *not* have network identifiability is that uniqueness holds whenever $G_c^1 \neq 0$ is internally contradictory. The intended statement is presumably that generic network identifiability *does* hold.

8.2 Example 2: reduced excitation is not proved by column deletion

When Q_1 is known, either component of

$$(G_1, G_1 Q_1) \tag{65}$$

can generically recover G_1, provided the known multiplier used is nonzero. That algebraic observation is correct.

The further conclusion that it suffices to estimate only a corresponding predictor column, and hence that either of two reduced spectral conditions is enough, relies on Remark 1's invalid deletion argument. The unestimated predictor columns can cancel changes in the selected column unless they are fixed or a conditional spectrum condition is proved. The example can be repaired by formulating two genuinely reduced prediction models and establishing their informativity separately.

8.3 Example 3: the twofold ambiguity is correct

The map is

$$a = \frac{G_1}{1 - G_1 G_2}, \qquad b = \frac{G_2}{1 - G_1 G_2}. \tag{66}$$

The involution

$$(G_1, G_2) \longmapsto \left(-\frac{1}{G_2}, -\frac{1}{G_1} \right) \tag{67}$$

leaves (a, b) unchanged. Writing $d = 1 - G_1 G_2$ gives

$$G_1 = ad, \qquad G_2 = bd, \qquad ab\, d^2 + d - 1 = 0. \tag{68}$$

For generic (a, b), the last equation has two distinct roots, so the generic fiber has two points. The Jacobian determinant is

$$\det J = \frac{1 + G_1 G_2}{(1 - G_1 G_2)^3}, \tag{69}$$

which is generically nonzero. The example therefore correctly illustrates generic local identifiability without generic global identifiability.

Two qualifications are needed. First, a noise-free scalar experiment does not automatically identify the entire two-column predictor: $y = G_c r$ makes (y, r) spectrally dependent, and full

predictor columns can change in cancelling directions unless the model class rules this out. What is directly identifiable from persistently exciting r is the closed-loop transfer G_c. Second, the counterexample to the conjecture cited in the paper uses a duplicated network together with equality constraints on repeated transfer functions. It therefore refutes the conjecture in that shared-parameter enlarged model; that modeling qualification should remain explicit.

8.4 Example 4: an exact Jacobian certificate

Example 4 contains eleven scalar unknown transfer-function values:

$$(G_{12}, G_{23}, G_{34}, G_{45}, G_{51}, Q_1, Q_2, Q_3, Q_4, Q_5, R_1). \tag{70}$$

Let the fifteen scalar entries of

$$F = R_y(I - G)^{-1}Q_r \tag{71}$$

be ordered row by row, and let J_F be the resulting 15×11 Jacobian. At the admissible point

$$(G_{12}, G_{23}, G_{34}, G_{45}, G_{51}) = \left(2, \frac{1}{3}, 3, \frac{1}{2}, 2\right), \tag{72}$$

$$(Q_1, Q_2, Q_3, Q_4, Q_5) = (1, 1, 3, 3, 1), \qquad R_1 = 2, \tag{73}$$

the cycle product is two, so $I - G$ is invertible. The 11×11 minor formed from the output-entry rows

$$F_{11}, F_{12}, F_{13}, F_{14}, F_{15}, F_{21}, F_{22}, F_{23}, F_{31}, F_{32}, F_{34} \tag{74}$$

has exact determinant

$$-16. \tag{75}$$

Hence rank $J_F = 11$ at this point, and the generic Jacobian rank is eleven. This independently certifies the manuscript's generic local-identifiability conclusion, without relying on the reported eleven-minute Gröbner computation.

The informativity conclusion is again generic rather than uniform. For example, if $R_1 = 0$, one of the measured-output rows can lose rank.

8.5 Example 5: explicit generic inverse

The tuple of unknowns in the manuscript contains Q_2 twice. The intended tuple appears to be

$$(G_{12}, Q_1, Q_2, R_1). \tag{76}$$

Set $g = G_{12}$ and $\rho = R_1$. The displayed map simplifies to

$$F = \begin{bmatrix} \rho g & \rho Q_1 & \rho(1 + gQ_2) & \rho g \\ 1 & 0 & Q_2 & 1 \end{bmatrix}. \tag{77}$$

It admits the explicit generic inverse

$$Q_2 = F_{2,3}, \tag{78}$$

$$\rho = F_{1,3} - F_{1,1}F_{2,3}, \tag{79}$$

$$g = \frac{F_{1,1}}{\rho}, \tag{80}$$

$$Q_1 = \frac{F_{1,2}}{\rho}. \tag{81}$$

Thus the example is generically globally identifiable, a stronger conclusion than the reported generic local identifiability. The first and fourth columns of (77) are identical, so removal of the last column indeed loses no identifiability information.

For informativity, zero signal components must be deleted from the target vector rather than retained as zero rows. A matrix with an actual zero row cannot have full row rank. The phrase "put the corresponding rows and columns equal to zero" should therefore be replaced by a precise deletion operation, consistent with the earlier parts of the paper.

9 Additional notation and logical corrections

The following points are smaller than the theorem-level issues but should be corrected.

- The known signal r is introduced with m_r components and later included in z with p_r components. Since it is the same signal, the dimensions should be identified explicitly.
- The notation in the displayed partition of S around equation (3) appears corrupted; the S_{ye} block should be checked.
- Later M-matrices sometimes use $+R_y$ where the original definition uses $-R$. A row sign change does not affect rank or minors, but the convention should be consistent.
- "Almost all G_c" should be interpreted intrinsically on the image closure, not with ambient Lebesgue measure when the image is lower-dimensional.
- On reducible constraint varieties, genericity and exceptional sets should be specified componentwise.
- The bibliography lists the year of reference [21] as 2913; this is evidently a typographical error.
- Computational claims should be accompanied by code, the exact polynomial system, variable order, monomial order, software version, and a machine-checkable certificate whenever runtime is reported as evidence.

10 Suggested replacement theorem package

A revised paper could be organized around the following four statements.

10.1 Actual and generic informativity

Theorem A. For the true parameter θ_0, positive definite Φ_u and full row rank of $\Pi(\cdot;\theta_0)$ almost everywhere imply positive definite Φ_z and hence predictor informativity.

Corollary A. If Π has full generic row rank on an irreducible well-posed parameter variety, then Theorem A applies outside a proper algebraic exceptional subset of true systems.

10.2 Generic rank

Theorem B. When all structural nonzeros of M are independent and P is generically nonsingular,

$$\operatorname{rank}_{\mathrm{g}} \Pi = \nu(\mathcal{B}(M)) - n. \tag{82}$$

A directed path corollary may be stated under a canonical or explicitly selected perfect matching of P.

10.3 Algebraic rank-drop locus

Theorem C. Let $X^\circ = \mathcal{V}(I_X) \setminus \mathcal{V}(\Delta)$ and define J_k by (37). Then the admissible rank-$< k$ locus is $\mathcal{V}(J_k) \cap X^\circ$ and its Zariski closure is $\mathcal{V}(J_k)$. If r is the componentwise generic rank, use J_r to describe the exceptional set below generic rank.

10.4 Generic local identifiability

Theorem D. For irreducible quasi-affine X and $Y = \overline{F(X)}$, generic local identifiability up to a finite ambiguity holds if and only if

$$\dim X = \dim Y. \tag{83}$$

Equivalently, at a generic smooth point, the differential of F has rank $\dim X$. The conclusion concerns generic image points, not every admissible image point.

These four statements preserve the main algebraic program while removing the present logical overreach.

11 Recommendation and revision priorities

The manuscript contains a potentially useful contribution, particularly in its attempt to accommodate partial measurements, direct terms, known modules, repeated parameters, and subnetwork boundary signals in one algebraic framework. The present theorem statements are nevertheless too broad for publication in their current form.

The highest-priority revisions are:

1. Replace Theorem 1 by separate fixed-system and generic-system statements, and propagate the generic qualifier through every example and conclusion.
2. Withdraw Theorem 2 in its present form. Replace it by the bipartite matching formula (31), with a separate directed-path corollary under an explicit matching hypothesis such as $P = I - G$.
3. Reformulate Theorem 3 using the full denominator product and saturation, and distinguish the exact quasi-affine locus from its affine closure.
4. Reformulate Theorem 4 using $Y = \overline{F(X)}$, irreducibility or componentwise hypotheses, and a generic rather than universal fiber statement.
5. Choose a rigorous interpretation of transfer-function variables and explain how pointwise identifiability at one frequency relates to identification of complete rational functions.
6. Replace the predictor-column deletion claims by a result that accounts for nuisance columns, for example through conditional spectra or the projected-kernel condition (55).
7. Correct Examples 1–5 and archive reproducible symbolic certificates.

After these changes, the Gröbner-basis and generic-fiber components could form a coherent and valuable paper. Without them, two of the central advertised conclusions—the unrestricted graph characterization and the generic-rank sufficiency claim for the actual experiment—are false.

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